

PCG Arena: A Web Platform for Blind A/B Testing of Procedural Content Generators

Antoni Czapski

January 2026

Abstract

Evaluating procedural content generators (PCGs) for video games poses significant methodological challenges, as player enjoyment is inherently subjective and difficult to quantify through automated metrics alone. This report presents **PCG Arena**, a web-based platform designed for blind A/B testing of Mario level generators through human preference data. The platform enables players to play two procedurally generated levels side-by-side in their browser, vote for their preferred level, and optionally tag gameplay characteristics. Votes are aggregated using the Glicko-2 rating system, which provides uncertainty-aware rankings with confidence intervals.

Key technical contributions include: (1) a browser-based Mario game engine faithfully ported from Java to TypeScript, (2) an adaptive matchmaking algorithm (AGIS) that optimizes for rapid rating convergence, (3) a builder profile system enabling researchers to submit and compete with their own generators, and (4) comprehensive research analytics including death heatmaps, player trajectories, and confusion matrices.

The platform is deployed at <https://pcg-arena.com>, featuring Google OAuth and email/password authentication, Twilio SendGrid integration for email verification, and Docker-based deployment on Google Cloud Platform. This work provides the research community with an accessible tool for empirically comparing procedural content generators through human evaluation.

Contents

1	Introduction	3
2	Background and Related Work	3
2.1	Procedural Content Generation	3
2.2	Mario AI Framework	4
2.3	Human Evaluation in PCG	4
2.4	Rating Systems for Pairwise Comparisons	4
3	System Architecture	5
3.1	High-Level Overview	5
3.2	Backend Architecture	5
3.3	Frontend Architecture	6
3.4	Database Design	6
4	Game Engine Implementation	7
4.1	TypeScript Port from Java	7
4.2	Physics and Gameplay	7
4.3	Level Format and Parsing	7

5	Battle and Voting System	8
5.1	Battle Flow	8
5.2	Voting Interface	9
5.3	Tagging System	9
6	Matchmaking Algorithm	9
6.1	Problem Definition	10
6.2	AGIS: Adaptive Glicko-Informed Selection	10
6.3	Generator Selection Weights	10
7	Rating System	11
7.1	Glicko-2 Algorithm	11
7.2	Rating Updates	11
7.3	Confidence Intervals	12
8	Leaderboard and Analytics	12
8.1	Leaderboard Display	12
8.2	Generator Statistics	13
8.3	Confusion Matrix	13
9	Authentication and User Management	15
9.1	Google OAuth Integration	15
9.2	Email/Password Authentication	16
9.3	Email Verification with Twilio SendGrid	16
10	Builder Profile	17
10.1	Generator Submission	17
10.2	Level Validation	17
10.3	Generator Management	18
11	Admin Dashboard	18
11.1	Platform Overview	18
11.2	Data Export	19
11.3	Research Analytics	20
12	Deployment and Infrastructure	22
12.1	Google Cloud Platform Setup	22
12.2	Docker Containerization	22
12.3	Domain and HTTPS Configuration	23
12.4	Monitoring and Alerts	23
13	Results and Discussion	24
13.1	Platform Statistics	24
13.2	Preliminary Generator Rankings	24
13.3	User Experience Insights	24
14	Conclusions and Future Work	25
14.1	Summary of Contributions	25
14.2	Limitations	25
14.3	Future Directions	25

1 Introduction

Procedural Content Generation (PCG) has become a cornerstone of modern game development, enabling the automatic creation of game levels, narratives, textures, and other assets [3]. However, evaluating the quality of procedurally generated content remains a fundamental challenge in the field. While automated metrics such as playability, diversity, and structural features provide valuable insights, they often fail to capture the subjective experience of human players—the ultimate consumers of generated content.

The Mario AI Championship’s Level Generation Track [2] established a precedent for human evaluation of PCG by having players experience and compare levels from different generators. However, such competitions typically rely on controlled laboratory settings with limited participants, making it difficult to gather statistically significant preference data at scale.

This project addresses these limitations by developing **PCG Arena**, a fully operational web platform that enables:

1. **Browser-Based Gameplay:** Players can experience Mario levels directly in their browser without downloading any software, removing the primary barrier to participation.
2. **Blind A/B Testing:** Each battle presents two levels from different generators without revealing their source, eliminating bias from generator reputation.
3. **Uncertainty-Aware Rankings:** The Glicko-2 rating system provides not only point estimates of generator quality but also confidence intervals that reflect the reliability of those estimates.
4. **Adaptive Matchmaking:** The AGIS (Adaptive Glicko-Informed Selection) algorithm prioritizes matchups that maximize information gain, ensuring efficient convergence of ratings.
5. **Research Analytics:** Comprehensive data collection enables analysis of player preferences, level difficulty, and engagement patterns.

The platform is live at <https://pcg-arena.com> and serves as both a practical tool for PCG researchers and a demonstration of modern web technologies for game-based research.

This report is structured as follows: Section 2 provides background on PCG and rating systems. Section 3 describes the system architecture. Sections 4–11 detail the core components. Section 12 covers infrastructure, and Section 13 presents preliminary results.

2 Background and Related Work

2.1 Procedural Content Generation

Procedural Content Generation refers to the algorithmic creation of game content with limited or indirect human input [3]. PCG techniques range from simple rule-based systems to sophisticated machine learning approaches, including:

- **Search-Based PCG:** Using evolutionary algorithms to optimize levels against fitness functions measuring playability, difficulty, and aesthetics.
- **Grammar-Based PCG:** Defining formal grammars that specify valid level structures and generating content through grammar expansion.
- **Machine Learning PCG (PCGML):** Training neural networks on existing human-designed content to learn implicit design patterns [3].

A fundamental challenge in PCG research is evaluation: how do we determine whether one generator produces “better” content than another? This question motivates the development of PCG Arena.

2.2 Mario AI Framework

The Mario AI Framework [2] provides a standardized environment for AI research in platformer games. Originally developed for the Mario AI Competition, it offers:

- A faithful recreation of Super Mario Bros. physics and mechanics
- A standardized ASCII level format for representing tile-based levels
- APIs for both playing agents and level generators
- Tools for level analysis and visualization

PCG Arena builds upon this framework, using its level format and adapting its physics engine for browser-based gameplay. The ASCII tilemap format represents levels as grids of characters, where each character denotes a specific tile type (ground, brick, enemy spawn, etc.).

2.3 Human Evaluation in PCG

Human evaluation remains the gold standard for assessing game content quality [2]. Common approaches include:

- **Absolute Rating:** Players rate individual levels on a numerical scale (e.g., 1–5 stars).
- **Pairwise Comparison:** Players compare two items and choose their preference.
- **Ranking:** Players order multiple items from best to worst.

Pairwise comparison, the method used by PCG Arena, offers several advantages: it reduces cognitive load compared to absolute rating, produces more consistent results across participants, and maps naturally to ranking algorithms like Elo and Glicko-2.

2.4 Rating Systems for Pairwise Comparisons

Rating systems for pairwise comparisons originated in chess and have been widely adopted for competitive games and recommendation systems. The Elo rating system, developed by Arpad Elo, models the probability of player A defeating player B based on their rating difference:

$$P(A \text{ wins}) = \frac{1}{1 + 10^{(R_B - R_A)/400}} \quad (1)$$

The Glicko-2 system [1] extends Elo by adding two additional parameters:

- **Rating Deviation (RD):** Measures uncertainty in the rating estimate. New or inactive players have high RD; active players converge to low RD.
- **Volatility (σ):** Captures the expected fluctuation in performance over time.

PCG Arena uses Glicko-2 because its uncertainty quantification directly addresses a key research need: knowing not just which generator is best, but how confident we can be in that ranking.

3 System Architecture

3.1 High-Level Overview

PCG Arena follows a classic three-tier web architecture with clear separation of concerns:

1. **Presentation Layer:** React/TypeScript frontend running in the user's browser
2. **Application Layer:** Python/FastAPI backend handling business logic and API endpoints
3. **Data Layer:** SQLite database persisting all platform state

Figure 1 illustrates the high-level system architecture. Users interact with the platform through a browser-based frontend that communicates with the backend via a RESTful API following the `arena/v0` protocol specification.

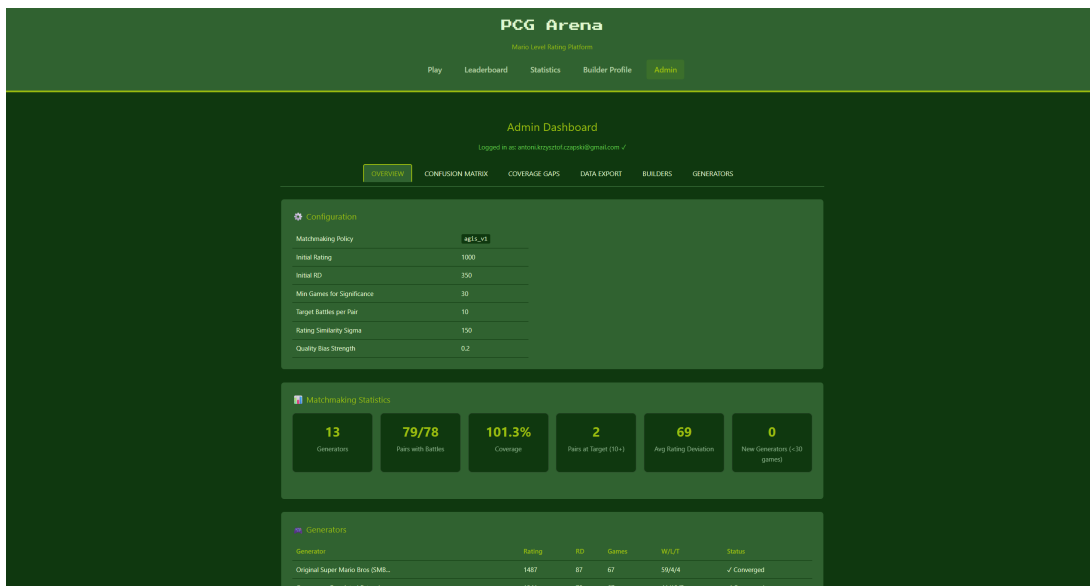


Figure 1: Admin overview showing system architecture and platform statistics.

3.2 Backend Architecture

The backend is implemented in Python using the FastAPI framework, chosen for its:

- Automatic OpenAPI documentation generation
- Native `async/await` support for concurrent request handling
- Built-in request validation via Pydantic models
- High performance comparable to Node.js and Go

Key backend modules include:

- `main.py`: FastAPI application with route definitions
- `auth.py`: Authentication logic (Google OAuth, email/password)
- `glicko2.py`: Glicko-2 rating system implementation
- `matchmaking.py`: AGIS matchmaking algorithm

- `builders.py`: Generator submission and management
- `stats.py`: Research analytics and data export

The backend runs in a Docker container, ensuring consistent deployment across environments.

3.3 Frontend Architecture

The frontend is built with React 18 and TypeScript 5.6, using Vite as the build tool. Key architectural decisions include:

- **Component-Based Design**: UI elements are reusable React components
- **Canvas-Based Rendering**: Game graphics use HTML5 Canvas for direct pixel manipulation
- **Type Safety**: TypeScript catches errors at compile time
- **Fast Development**: Vite provides sub-second hot module replacement

The frontend directory structure separates concerns:

```
frontend/src/
  api/ # Backend communication
  engine/ # Game engine (ported from Java)
  components/ # React UI components
  pages/ # Full page components
  contexts/ # React state management
```

3.4 Database Design

PCG Arena uses SQLite for data persistence, chosen for:

- Single-file storage simplifying backups and deployment
- ACID transactions ensuring data consistency
- Zero configuration and no separate database service
- Sufficient performance for the expected load

The database schema comprises 17 tables across 17 migrations:

Table	Purpose
<code>generators</code>	PCG algorithm metadata with owner reference
<code>levels</code>	ASCII tilemap levels with content hash
<code>battles</code>	Comparison instances pairing two levels
<code>votes</code>	User outcomes with telemetry data
<code>ratings</code>	Glicko-2 ratings per generator
<code>rating_events</code>	Audit log of rating changes
<code>users</code>	Authenticated user accounts
<code>user_sessions</code>	Active login sessions
<code>play_trajectories</code>	Player position and death data

Table 1: Key database tables in PCG Arena.

4 Game Engine Implementation

4.1 TypeScript Port from Java

A critical component of PCG Arena is the browser-based Mario game engine, which was ported from the Java-based Mario AI Framework to TypeScript. This port required careful attention to maintain gameplay fidelity while adapting to browser constraints.

The porting process involved:

1. **Class Structure Translation:** Java classes were converted to TypeScript classes, preserving inheritance hierarchies and encapsulation.
2. **Physics Constants Preservation:** All physics constants were kept identical to ensure matching gameplay feel.
3. **Rendering System Adaptation:** Java’s Swing/AWT rendering was replaced with HTML5 Canvas.
4. **Input System Rewrite:** Keyboard input handling was reimplemented for browser events.

Key engine modules include:

```
engine/  
MarioGame.ts # Game loop (requestAnimationFrame)  
MarioWorld.ts # World state and game logic  
MarioLevel.ts # Level parsing from ASCII  
MarioSprite.ts # Base sprite class  
sprites/ # Mario, enemies, items  
graphics/ # Camera, rendering  
effects/ # Visual effects (particles)
```

4.2 Physics and Gameplay

The game engine preserves exact physics constants from the Java version to ensure identical gameplay:

Parameter	Value
Ground Inertia	0.89
Air Inertia	0.89
Gravity	3.0
Jump Velocity	−1.9
Walk Speed	0.6
Run Speed	1.2

Table 2: Core physics constants in the Mario engine.

The game loop runs at 30 ticks per second (matching the original), with rendering at 60 FPS using `requestAnimationFrame`. Physics updates are deterministic, ensuring consistent gameplay across sessions.

4.3 Level Format and Parsing

Levels are represented as ASCII tilemaps, where each character corresponds to a tile type:

```

----- (empty space)
-----S--- (S = breakable brick)
---E---Q----- (E = enemy, Q = question block)
##### (# = ground)

```

The level parser (`MarioLevel.ts`) processes these tilemaps, converting characters to internal tile representations and spawning sprite entities for enemies and items. Validation rules ensure levels are playable:

- Minimum dimensions: 50×16 tiles
- Maximum dimensions: 500×16 tiles
- Required: Mario spawn point and exit flag
- Prohibited: Unreachable exits, instant-death spawns

5 Battle and Voting System

5.1 Battle Flow

The core user experience of PCG Arena is the “battle”—a blind A/B test where players compare two levels from different generators. The battle flow proceeds as follows:

1. **Battle Request:** The frontend requests a new battle from `POST /v1/battles:next`.
2. **Matchmaking:** The AGIS algorithm selects two generators and one random level from each.
3. **Gameplay:** The player plays both levels sequentially (left then right).
4. **Voting:** After playing both, the player votes for their preference.
5. **Rating Update:** The backend updates Glicko-2 ratings atomically.

Figure 2 shows the main gameplay interface where players experience levels side-by-side.



Figure 2: Main gameplay view showing the active level with game controls and timer.

5.2 Voting Interface

After completing both levels, players are presented with the voting interface (Figure 3). The interface displays:

- Static visualizations of both complete levels for reference
- Vote buttons: “Left is Better,” “Right is Better,” “Tie,” “Skip”
- Optional tagging interface for additional feedback



Figure 3: Voting panel showing level previews and vote options after gameplay.

The “Skip” option allows players to abstain from voting if they encountered technical issues or feel unable to judge. Skipped battles do not affect ratings.

5.3 Tagging System

In addition to the binary preference vote, players can optionally tag each level with descriptive labels. Tags capture qualitative aspects that the vote alone cannot express:

Category	Tags
Difficulty	too-easy, just-right, too-hard
Design	creative, boring, frustrating, fun
Structure	linear, exploration, puzzle-like
Issues	broken, impossible, unplayable

Table 3: Available tags for level characterization.

Tag data enables richer analysis, such as correlating perceived difficulty with win rates or identifying generators that produce consistently “frustrating” levels.

6 Matchmaking Algorithm

6.1 Problem Definition

The matchmaking problem in PCG Arena is: given n generators with current ratings and uncertainties, select a pair for the next battle that maximizes information gain toward accurate final rankings.

Naive approaches fail:

- **Uniform Random:** Wastes battles on already-converged pairs.
- **Most Uncertain First:** May neglect head-to-head comparisons between similar generators.
- **Round-Robin:** Ignores rating information; pairs dissimilar generators.

The ideal matchmaking algorithm should:

1. Prioritize generators with high uncertainty (new or rarely played)
2. Prefer matchups between generators of similar skill levels
3. Ensure all pairs get sufficient battles for the confusion matrix
4. Converge quickly to stable rankings

6.2 AGIS: Adaptive Glicko-Informed Selection

We developed **AGIS (Adaptive Glicko-Informed Selection)**, a matchmaking algorithm that uses Glicko-2 rating information to optimize battle selection. AGIS operates in two stages:

Stage 1: First Generator Selection

Each generator i receives a selection weight w_i based on:

$$w_i = (1 + \text{RD}_{\text{norm},i})^2 \cdot \text{boost}(\text{games}_i) \cdot (1 + \alpha \cdot \text{quality}_i) \quad (2)$$

where RD_{norm} is the normalized rating deviation (high = uncertain), boost amplifies weight for generators with few games, and quality provides slight bias toward higher-rated generators after convergence.

Stage 2: Second Generator Selection

Given the first generator g_1 , the second generator is selected with weight:

$$w_{1,j} = \alpha \cdot S(g_1, g_j) + \beta \cdot U(g_j) + \gamma \cdot C(g_1, g_j) \quad (3)$$

where:

- $S(g_1, g_j)$: Rating similarity (Gaussian decay from rating difference)
- $U(g_j)$: Uncertainty factor (based on g_j 's RD)
- $C(g_1, g_j)$: Coverage factor (inverse of pair's battle count)

6.3 Generator Selection Weights

The configurable parameters of AGIS (with defaults) are:

Parameter	Description	Default
AGIS_MIN_GAMES	Games before “converged”	30
AGIS_TARGET_BATTLES_PER_PAIR	Minimum battles per pair	10
AGIS_RATING_SIGMA	Rating similarity std dev	150.0
AGIS_QUALITY_BIAS	Quality bias strength	0.2
α	Rating similarity weight	0.5
β	Uncertainty weight	0.3
γ	Coverage weight	0.2

Table 4: AGIS algorithm parameters.

7 Rating System

7.1 Glicko-2 Algorithm

PCG Arena uses the Glicko-2 rating system [1], which maintains three parameters for each generator:

- **Rating** (μ): The skill estimate, displayed on a scale centered at 1000.
- **Rating Deviation (RD, ϕ)**: The uncertainty in the rating. High RD means unreliable rating.
- **Volatility** (σ): Expected fluctuation in performance over time.

The expected score when generator A plays against generator B is:

$$E(\mu_A, \mu_B, \phi_B) = \frac{1}{1 + \exp(-g(\phi_B)(\mu_A - \mu_B))} \quad (4)$$

where $g(\phi) = \frac{1}{\sqrt{1+3\phi^2/\pi^2}}$ reduces the impact of uncertain opponents.

7.2 Rating Updates

After each battle, the winning generator receives a score of 1, the loser 0, and ties result in 0.5 for both. The rating update proceeds through several steps:

1. Convert ratings to Glicko-2 scale (centered at 0, scaled by 173.7178)
2. Compute variance v based on opponent’s RD
3. Compute delta Δ : actual minus expected score
4. Update volatility σ' using iterative algorithm
5. Update RD: $\phi' = 1/\sqrt{1/\phi^2 + 1/v}$
6. Update rating: $\mu' = \mu + \phi'^2 \cdot g(\phi_B) \cdot (s - E)$
7. Convert back to display scale

Rating updates are performed atomically within a database transaction to ensure consistency.

7.3 Confidence Intervals

A key advantage of Glicko-2 is that RD provides natural confidence intervals. The 95% confidence interval for a generator’s true rating is approximately:

$$CI_{95\%} = [\mu - 1.96 \cdot RD, \mu + 1.96 \cdot RD] \quad (5)$$

This enables meaningful statements like “Generator A is rated 1200 ± 50 with 95% confidence,” which is essential for research applications.

Rating bounds are enforced:

- Minimum rating: 100 (prevents extreme negative outliers)
- Maximum rating: 3000 (theoretical maximum)
- Minimum RD: 30 (very confident)
- Maximum RD: 350 (new/inactive)

8 Leaderboard and Analytics

8.1 Leaderboard Display

The public leaderboard (Figure 4) displays all active generators ranked by their Glicko-2 rating. For each generator, the display shows:

- Rank position
- Generator name and version
- Current rating with $\pm$ RD uncertainty
- Number of battles played
- Win/loss/tie record

Rank	Generator	Rating	Games	W	L	T
1	Original Super Mario Bros. (SMB1) Levels	1457	67	59	4	4
2	Randomly-Generated Extensions (GRG) Generator	1241	67	41	19	7
3	Notch (Infinite Mario Bros) Generator	1157	60	36	17	7
4	Grammatical Evolution (GE) Generator	1072	65	32	23	10
5	Random Level Generator	1060	65	34	26	5
6	MarinGAN (DCGAN + CNN-ES Latent-Space Generator)	1036	69	32	29	8
7	MarinGPT (Text2Level via GPT-2)	1012	76	42	30	6
8	Parameterized Notch (Randomized Params) Generator	952	70	24	34	12
9	Pattern-based Generator (Weighted Pattern Count Fitness)	926	73	19	44	10
10	Pattern-based Generator (Pattern Occurrence Fitness)	914	59	15	33	6
11	MarinDiffusion (Text-to-Level Diffusion)	859	69	20	35	14
12	Parameterized Notch Generator	840	66	12	42	13
13	Pattern-based Generator (Pattern Count Fitness)	790	69	11	44	14

Figure 4: Public leaderboard showing generator rankings with Glicko-2 ratings and uncertainty intervals.

The leaderboard updates in real-time as votes are submitted, allowing users to see the immediate impact of their participation.

8.2 Generator Statistics

Clicking on a generator in the leaderboard reveals detailed statistics (Figure 5):

- Rating history over time
- Win rate against each opponent
- Level gallery with performance metrics per level
- Tag distribution (what players say about this generator’s levels)
- Average play statistics (deaths, completion rate, time)

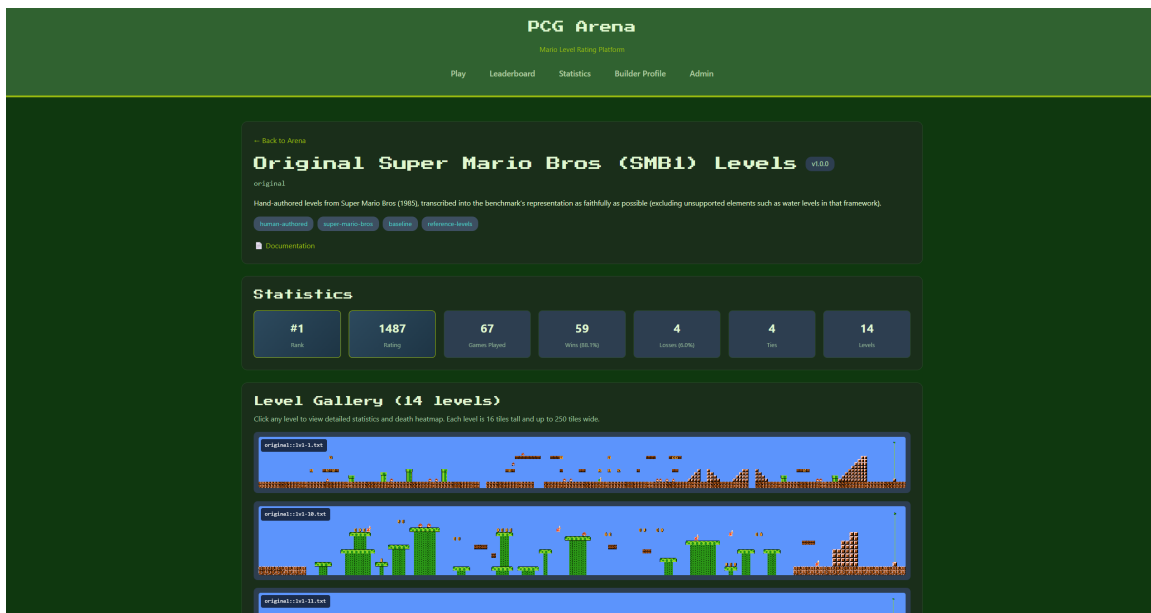


Figure 5: Generator detail page showing level gallery and statistics.

8.3 Confusion Matrix

The confusion matrix (Figure 6) provides a comprehensive view of head-to-head performance between all generator pairs. Each cell shows the win rate when the row generator faces the column generator.

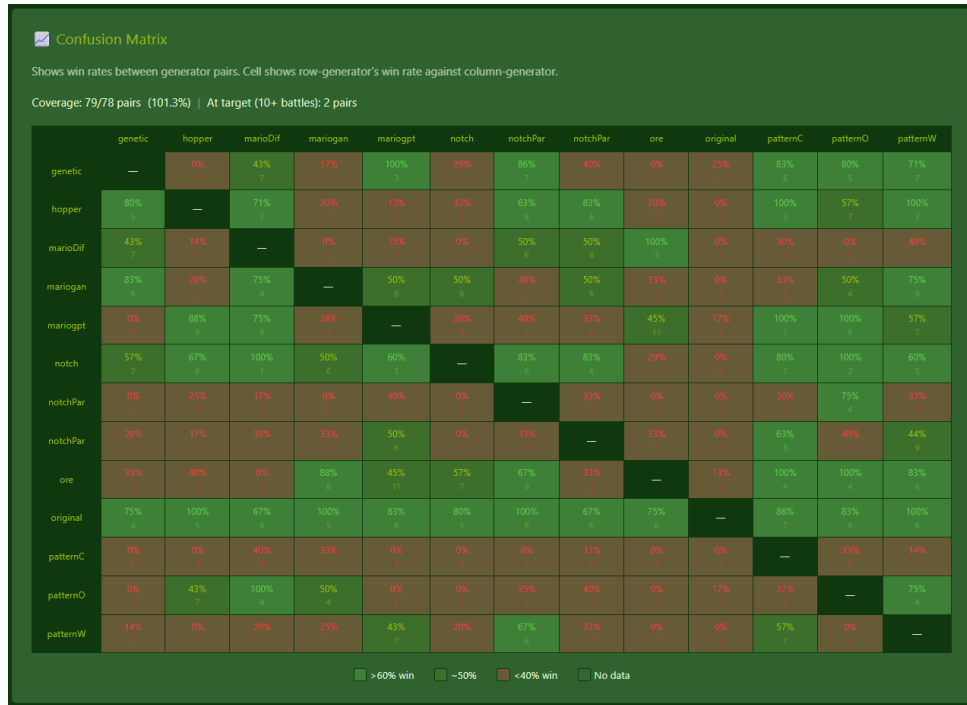


Figure 6: Confusion matrix showing pairwise win rates between generators.

The matrix is particularly useful for:

- Identifying rock-paper-scissors relationships (A beats B, B beats C, C beats A)
- Detecting generators with inconsistent performance
- Ensuring sufficient coverage—empty or low-count cells indicate gaps

Coverage gaps (Figure 7) are highlighted to guide future data collection.

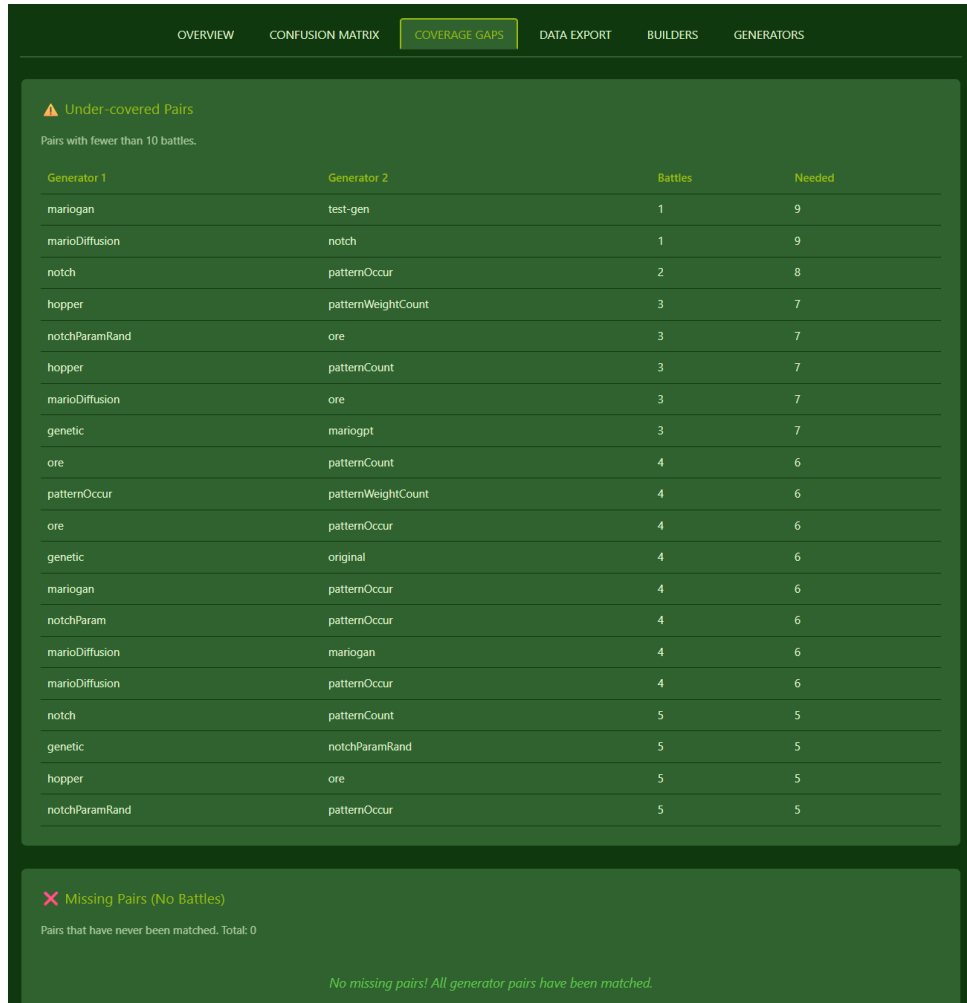


Figure 7: Coverage analysis showing pairs with insufficient battle data.

9 Authentication and User Management

9.1 Google OAuth Integration

PCG Arena supports Google OAuth 2.0 for quick sign-in, leveraging Google’s secure identity verification. The flow proceeds as:

1. User clicks “Sign In with Google”
2. Browser opens Google OAuth popup
3. User authenticates with Google credentials
4. Google returns an ID token (JWT) to the frontend
5. Frontend sends token to `POST /v1/auth/google`
6. Backend verifies token with Google’s servers
7. Backend creates or retrieves user record
8. Backend issues session cookie (`arena_session`)
9. User is redirected to Builder Profile

Google OAuth users are automatically marked as email-verified (Google handles this), enabling immediate access to submission features.

9.2 Email/Password Authentication

For users who prefer not to use Google, PCG Arena supports traditional email/password authentication:

Registration:

1. User provides email, password, and display name
2. Backend validates email format and password strength
3. Password is hashed using bcrypt (work factor 12)
4. User record created with `is_email_verified = 0`
5. Verification email sent (see Section 9.3)

Login:

1. User provides email and password
2. Backend retrieves user record by email
3. Password verified against stored hash
4. Session cookie issued on successful verification

Security measures include:

- Password minimum length: 8 characters
- Session tokens: 256-bit cryptographically random
- Session expiry: 7 days (configurable)
- HTTP-only cookies prevent XSS token theft

9.3 Email Verification with Twilio SendGrid

Email verification ensures that users control the email addresses they register with. PCG Arena uses Twilio SendGrid for email delivery:

1. On registration, generate a random verification token (64 hex characters)
2. Store token in `email_verifications` table with 24-hour expiry
3. Send verification email via SendGrid API containing link:
`https://pcg-arena.com/verify-email?token=<token>`
4. User clicks link, frontend calls `POST /v1/auth/verify-email`
5. Backend verifies token validity and expiry
6. User marked as verified; token deleted

Password reset follows a similar flow, using separate tokens stored in `password_resets`. Reset tokens expire after 1 hour for security.

10 Builder Profile

10.1 Generator Submission

The Builder Profile (Figure 8) enables authenticated researchers to submit their own generators to compete on the leaderboard. The submission process:

1. User navigates to Builder Profile (requires authentication)
2. Fills out generator metadata: name, version, description, tags, URL
3. Uploads a ZIP file containing level files (ASCII tilemaps)
4. Backend validates all levels (see Section 10.2)
5. Generator and levels are inserted into the database
6. Initial Glicko-2 rating assigned (1000 ± 350)
7. Generator appears on leaderboard immediately

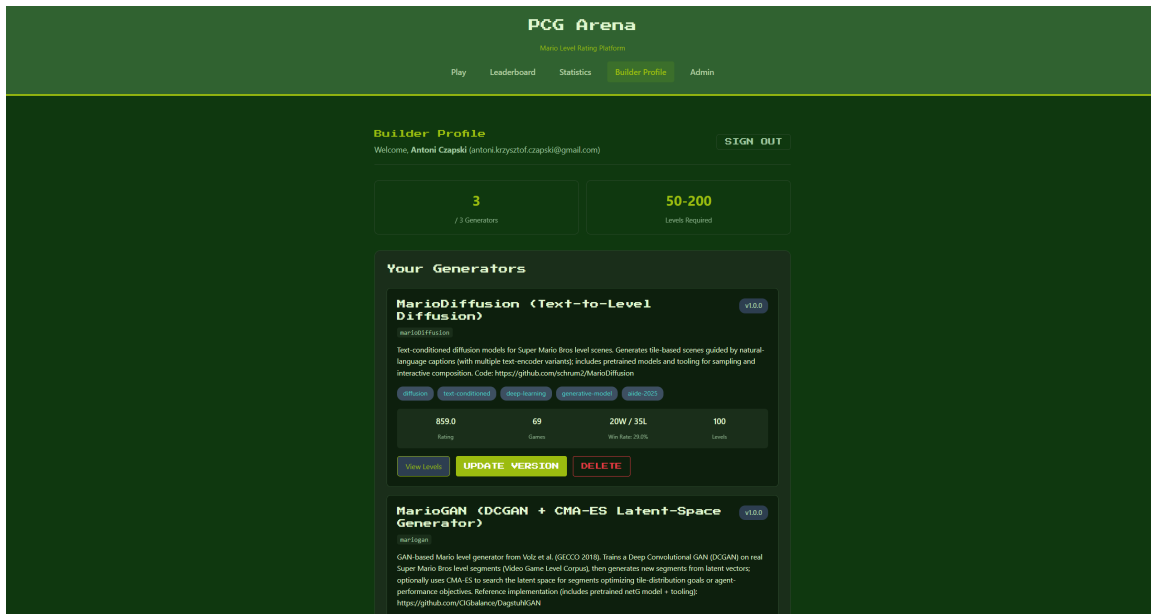


Figure 8: Builder Profile page for managing submitted generators.

Constraints on submissions:

- Maximum 3 generators per user (prevents flooding)
- 50–200 levels per generator (ensures variety without overwhelming)
- Each level must pass validation checks

10.2 Level Validation

Every submitted level undergoes validation to ensure playability:

1. **Format Check:** File must be valid ASCII, correct dimensions
2. **Character Validation:** Only recognized tile characters allowed

3. **Spawn/Exit Check:** Level must have Mario spawn point and reachable exit
4. **Deduplication:** Content hash computed; duplicates rejected
5. **Playability Heuristics:** Basic reachability analysis

Validation errors return detailed feedback indicating which levels failed and why, enabling researchers to fix issues before resubmitting.

10.3 Generator Management

After submission, researchers can manage their generators:

Update Version:

- Upload new level set while preserving rating
- Version number incremented
- Old levels replaced with new ones
- Rating history maintained for continuity

Delete Generator:

- Generators with battles: soft delete (marked inactive, preserved for historical data)
- Generators without battles: hard delete (completely removed)
- Soft-deleted generators don't appear in matchmaking but remain in historical rankings

This design ensures data integrity—votes referencing deleted generators remain valid for research analysis.

11 Admin Dashboard

11.1 Platform Overview

The Admin Dashboard (Figure 9) provides platform administrators with comprehensive oversight and management capabilities. Access is restricted to users whose email addresses are listed in the `ARENA_ADMIN_EMAILS` configuration.

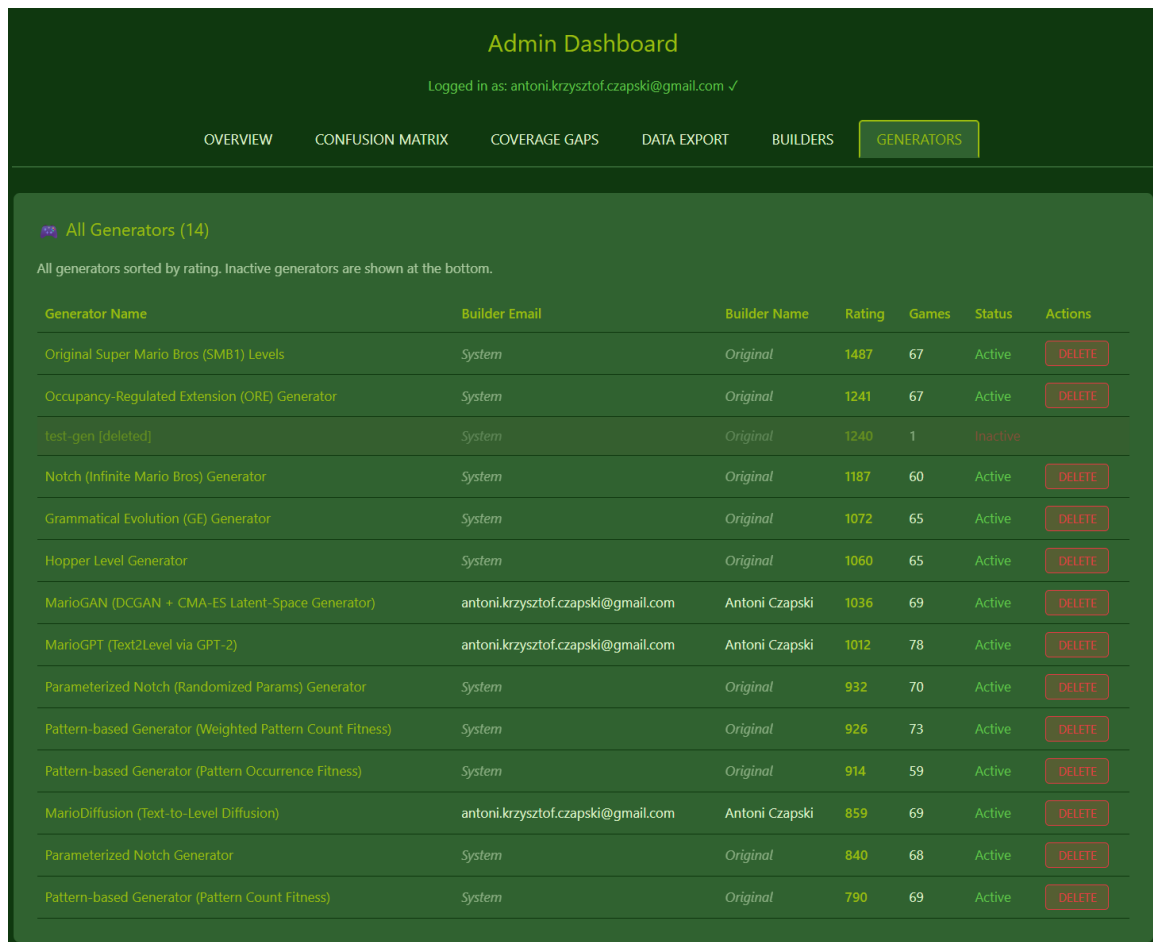


Figure 9: Admin Dashboard showing platform statistics and management tools.

Key metrics displayed:

- Total battles completed and pending
- Total votes and vote distribution (left/right/tie/skip)
- Active generators and users
- Recent activity timeline
- System health indicators

Administrative actions available:

- View and ban builder accounts
- Delete inappropriate generators
- Monitor matchmaking statistics
- Trigger manual data exports

11.2 Data Export

PCG Arena provides comprehensive data export for research purposes (Figure 10). Export endpoints (admin-only) include:

Endpoint	Data
/v1/admin/export/votes	All votes with telemetry and tags
/v1/admin/export/trajectories	Player position traces
/v1/admin/export/level-stats	Aggregate per-level metrics
/v1/admin/export/player-profiles	Anonymous player IDs and sessions

Table 5: Available data export endpoints.

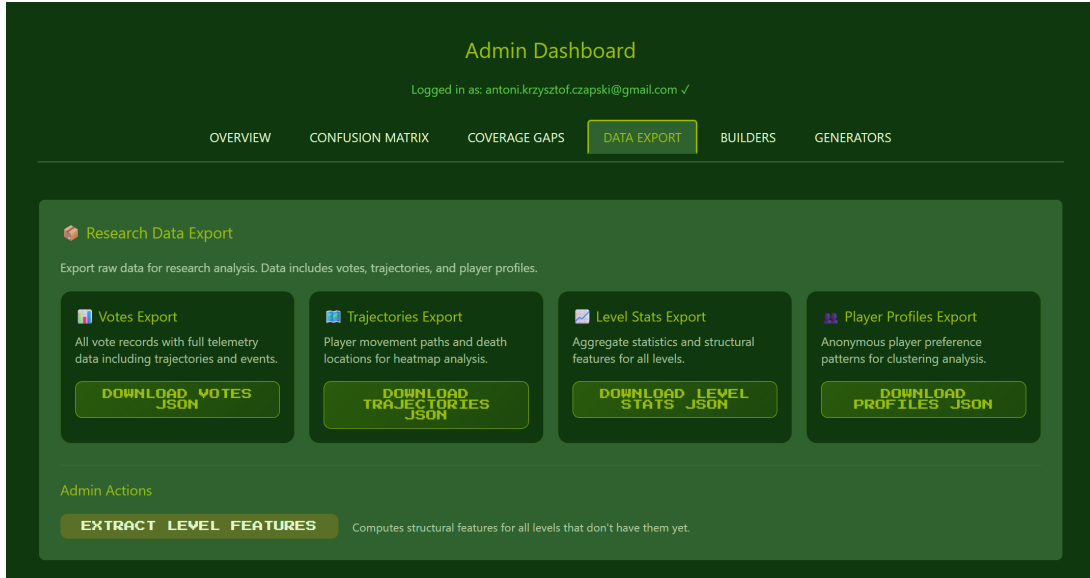


Figure 10: Data export interface for downloading research datasets.

Data is exported in JSON and CSV formats suitable for analysis with standard tools (Python/pandas, R, etc.).

11.3 Research Analytics

The platform collects rich telemetry enabling advanced research analytics:

Level Statistics (Figure 11):

- Completion rate: percentage of players who finish
- Death rate: average deaths per attempt
- Average play time
- Win rate when featured in battles

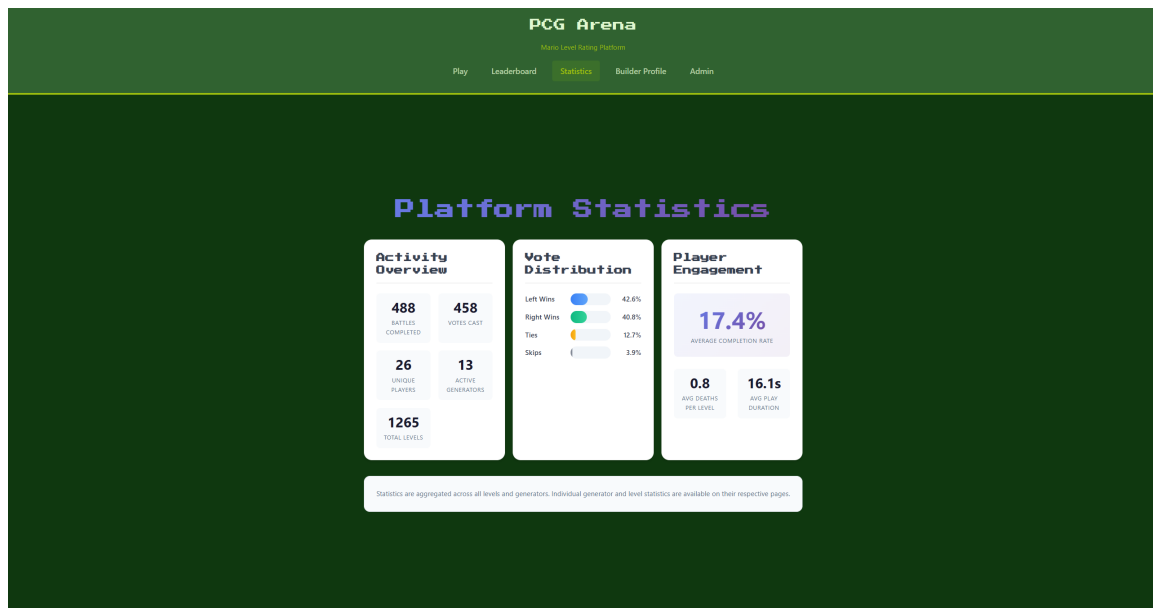


Figure 11: Statistics page showing detailed level and generator analytics.

Death Heatmaps:

- Aggregate death locations across all plays of a level
- Reveals difficulty spikes and design issues
- Exportable for external visualization

Detailed Level View (Figure 12):

- Full level visualization with overlay options
- Play trajectory visualization
- Per-tile analysis

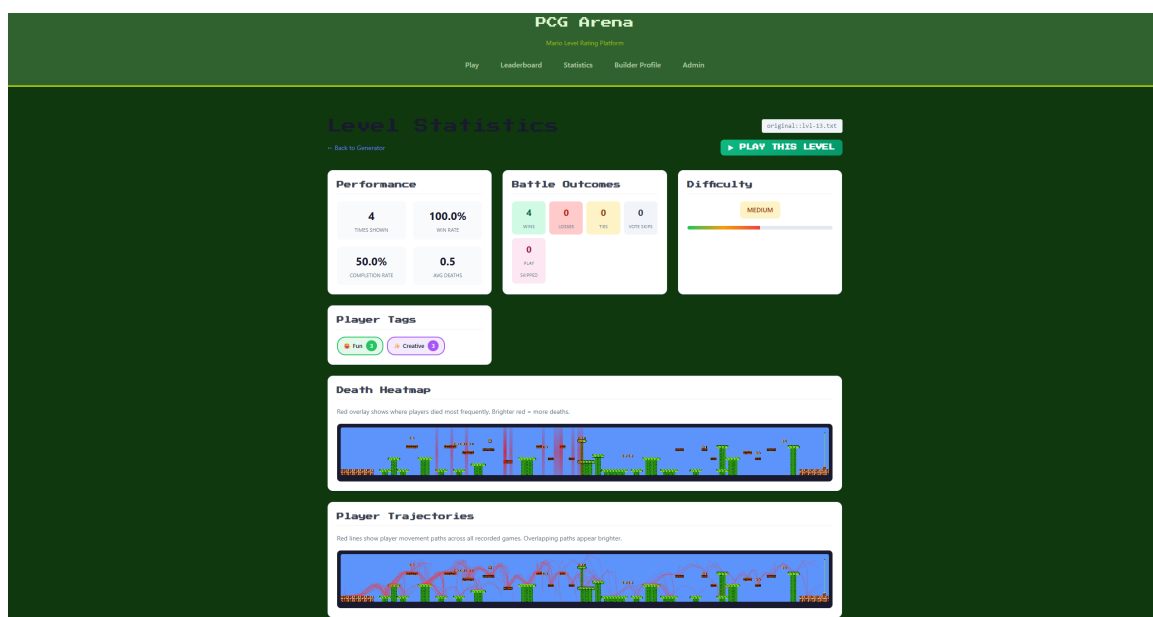


Figure 12: Detailed level view with analytics overlays.

12 Deployment and Infrastructure

12.1 Google Cloud Platform Setup

PCG Arena is deployed on Google Cloud Platform (GCP) using a single Compute Engine virtual machine. This setup balances cost-effectiveness with reliability for a research platform:

VM Specifications:

- Machine type: **e2-micro** (free tier eligible)
- Region: **us-central1** (low latency for most users)
- Boot disk: Ubuntu 22.04 LTS, 10 GB standard persistent disk
- Static external IP: reserved for DNS stability

Firewall Configuration:

- Port 80 (HTTP): Allowed for redirect to HTTPS
- Port 443 (HTTPS): Allowed for all traffic
- Port 8080 (backend): **NOT** public; backend binds to localhost

The architecture ensures that all public traffic goes through the Caddy reverse proxy, which handles TLS termination and routes requests appropriately.

12.2 Docker Containerization

The backend runs in a Docker container, providing:

- Consistent environment across development and production
- Easy deployment via `docker compose up -d --build`
- Isolated dependencies (Python packages)
- Simple rollback by switching image tags

The `docker-compose.yml` configuration:

```
services:
  backend:
    build: ./backend
    ports:
      - "127.0.0.1:8080:8080" # Localhost only
    volumes:
      - ./db/local:/data
      - ./db/migrations:/migrations
      - ./db/seed:/seed
    environment:
      - ARENA_DB_PATH=/data/arena.sqlite
```

Database files are bind-mounted from the host, ensuring persistence across container rebuilds.

12.3 Domain and HTTPS Configuration

The platform is accessible at <https://pcg-arena.com>, configured as follows:

Domain Registration:

- Domain: `pcg-arena.com` (registered via Namecheap)
- DNS A records: `@` and `www` point to VM's static IP

HTTPS via Caddy:

- Automatic TLS certificate provisioning (Let's Encrypt)
- Automatic certificate renewal
- HTTP to HTTPS redirect
- Reverse proxy to backend (`localhost:8080`)
- Static file serving for frontend build

Caddyfile configuration:

```
pcg-arena.com, www.pcg-arena.com {
  root * /var/www/pcg-arena/frontend/dist
  file_server

  @api path /v1/* /health
  handle @api {
    reverse_proxy localhost:8080
  }

  try_files {path} /index.html
}
```

12.4 Monitoring and Alerts

GCP provides built-in monitoring capabilities:

- **Uptime Checks:** HTTP(S) probe to `/health` endpoint every 60 seconds
- **Alerting Policies:** Email notification when health check fails
- **VM Metrics:** CPU, memory, disk usage monitoring
- **Logging:** Application logs accessible via Cloud Console

Automated backups run daily via cron job:

```
0 3 * * * /home/user/pcg-arena/backend/scripts/backup.sh
```

Backups are stored in `/backups/` with 7-day retention, enabling recovery from data corruption or accidental deletion.

13 Results and Discussion

13.1 Platform Statistics

As a newly deployed platform, PCG Arena is in its initial data collection phase. The platform infrastructure has been fully validated through:

- End-to-end testing of battle flow (request → play → vote → rating update)
- Automated demo scripts running 10+ battles with random votes
- Manual testing across multiple browsers (Chrome, Firefox, Safari, Edge)
- Load testing confirming responsiveness under concurrent users

Current seeded generators include:

- **Notch:** Original Super Mario Bros. level design style
- **Hopper:** Pattern-based generation
- **Genetic:** Evolutionary algorithm-based generation

Each generator is seeded with 10+ levels for initial testing, with access to larger level sets (1000+ levels per generator) from the Mario AI Framework available for future expansion.

13.2 Preliminary Generator Rankings

Initial rankings reflect the starting state (all generators at 1000 ± 350) plus any early votes. As more battles are completed, we expect to observe:

1. **RD Convergence:** Rating deviations should decrease from 350 toward 30–100 as confidence increases.
2. **Rating Separation:** Distinct quality differences between generators should emerge.
3. **Stable Rankings:** After sufficient battles, rank order should stabilize.

The Glicko-2 system’s uncertainty quantification enables statistical tests for whether observed differences are significant.

13.3 User Experience Insights

Based on initial testing and informal feedback:

Strengths:

- Browser-based gameplay removes download barrier—users can participate immediately
- Side-by-side comparison format is intuitive
- Tagging system captures nuanced feedback beyond simple preferences
- Real-time leaderboard updates provide immediate gratification

Areas for Improvement:

- Mobile/touch support not yet implemented (desktop-only)
- Level difficulty can vary significantly, affecting fair comparison
- Some users find 30-second levels too short for assessment
- Tutorial for first-time users would improve onboarding

These insights inform future development priorities.

14 Conclusions and Future Work

14.1 Summary of Contributions

This project presents **PCG Arena**, a complete web platform for blind A/B testing of procedural content generators through human preference data. Key contributions include:

1. **Browser-Based Mario Engine:** A faithful TypeScript port of the Mario AI Framework game engine, enabling gameplay without downloads.
2. **Uncertainty-Aware Rating System:** Implementation of Glicko-2 providing not just rankings but confidence intervals essential for research validity.
3. **Adaptive Matchmaking (AGIS):** A novel algorithm optimizing battle selection for rapid rating convergence while ensuring coverage.
4. **Research Infrastructure:** Comprehensive telemetry, data export, and analytics tools enabling research-grade data collection.
5. **Open Platform:** Builder Profile system allowing researchers to submit and compete with their own generators.
6. **Production Deployment:** Live platform at <https://pcg-arena.com> with proper security, authentication, and monitoring.

14.2 Limitations

Several limitations should be acknowledged:

- **Platform-Specific:** Currently limited to Mario-style platformer levels; generalizing to other game genres requires significant adaptation.
- **Desktop-Only:** No touch/mobile support limits the potential participant pool.
- **Selection Bias:** Self-selected participants may not represent general player preferences.
- **Level Length:** Fixed 30-second gameplay may not capture all relevant differences.
- **Single Metric:** Pairwise preference conflates multiple quality dimensions (fun, difficulty, aesthetics).

14.3 Future Directions

Several directions for future work emerge from this project:

Short-term Improvements:

- Mobile/touch control support
- Tutorial and onboarding flow
- Social features (sharing results, challenging friends)
- Expanded tag vocabulary based on user feedback

Research Extensions:

- Multi-armed bandit approaches for matchmaking
- Machine learning models predicting preference from level features

- Player clustering to identify preference types
- Cross-generator level morphing experiments

Platform Generalization:

- Support for other game genres (puzzle games, roguelikes)
- Pluggable game engine architecture
- API for programmatic generator submission
- Federation with other PCG evaluation platforms

PCG Arena represents a step toward democratizing human evaluation in PCG research, providing the community with an accessible tool for empirically grounded generator comparison.

References

- [1] M. E. Glickman, “Example of the Glicko-2 system,” <http://www.glicko.net/glicko/glicko2.pdf>, 2012.
- [2] A. Khalifa, M. C. Green, D. Perez-Liebana, and J. Togelius, “General Video Game Level Generation,” *Proceedings of the Genetic and Evolutionary Computation Conference*, 2019.
- [3] A. Summerville, S. Snodgrass, M. Guzdial, C. Holmgård, A. K. Hoover, A. Isaksen, A. Nealen, and J. Togelius, “Procedural Content Generation via Machine Learning (PCGML),” *IEEE Transactions on Games*, vol. 10, no. 3, pp. 257–270, 2018.